SILTHAUL · DESIGN DECISION RECORD

SiltHaul STEM check fixes

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DOCUMENT CONTROL

Document ID	SLH-DDR-006	Version	0.1
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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/silthaul		
Source	docs/decisions/0006-stem-check-fixes.md @ 9347c2b		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-03	Amish Chadha	Two STEM fixes decided under Amish's standing instruction of 2026-10-23

0006: STEM check fixes

- **Date:** 2026-11-03
- **Status:** Decided under Amish's standing instruction of 2026-10-23: when a STEM check row fails, propose and implement the best fix for the design and its constructability in the same session; a fix that changes the working principle, pitch or scope goes to Amish instead. Neither fix below changes the working principle, the pitch or the scope.

Context

The STEM check (SLH-STM-001) of options A, B and C found two details that failed their first screen. Option A had no failing row.

Options considered

Table 1. Fixes considered.

#	Option	Failure	Fixes considered	Chosen
1	B	Boots on wet mud give a friction coefficient of about 0.25 (estimate); a 189 N pull needs 0.28	(a) staked foot boards with heel cleats; (b) a longer rope and more haulers; (c) a post to lean on	(a): keeps the principle, uses stock timber and four 25 mm stakes
2	C	A 120 mm stroke on a 250 mm flat EPDM diaphragm is 15 % strain each way and does not fit the 25 mm dish below the plane	(a) stroke 40 mm; (b) a convoluted diaphragm; (c) a bigger dish	(a): strain 1.7 %; the cover cavity is cut to 49 mm. Output falls to 0.44 m3/h with rest

Decision

Both fixes are made in the models (`cad/src/opt_b.py`, `cad/src/opt_c.py`, `PARAMS["pump"]["stroke_chamber"]` is 40 mm) and the checks rerun (`docs/04-calcs/stem_check.py`).

Consequences

- Option B's foot boards (two 40 x 200 x 1200 mm boards, heel cleats, coach screws, four stakes) are in its D3 model; they are not priced.
- Option C's R1 ratio falls to 0.49, which is part of why it scores lowest (SLH-OPT-001).
- Option A is not touched.